Literature Review

Federal Workforce Intelligence & Retention Analytics

1. Establishing Context

Previous studies on Human Resource Analytics and Workforce Planning have shown how data analysis can help organizations understand workforce composition, employee turnover, hiring trends, and staffing needs. These studies highlight the importance of using data-driven insights to improve workforce management and decision-making.

2. Identifying the Research Gap

Although workforce analytics has been widely studied, this project focuses on large-scale U.S. federal workforce data from the Office of Personnel Management (OPM). It aims to analyze employment, accession, and separation records together to understand workforce changes over time and identify potential staffing challenges across agencies and occupational groups.

3. Research Methodology

The project will use Python to process and clean large TXT files, SQL Server to build a structured data warehouse, and Power BI or Tableau to create interactive dashboards. The analysis will focus on workforce size, hiring and separation trends, net workforce change, and patterns that may support workforce planning.

4. Avoiding Duplication and Building a Conceptual Framework

Existing research will guide the selection of analytical methods and performance indicators. The project will apply these methods to OPM data and examine how workforce inflows and outflows relate to changes in workforce size.

5. Expected Contribution

The project aims to transform large-scale federal workforce data into meaningful insights that help users monitor employment trends, compare hiring and separation patterns, and support evidence-based workforce planning and resource allocation.

Conceptual Framework

OPM Data
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Data Cleaning and Integration
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Workforce Analysis
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Interactive Dashboards
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Workforce Planning Insights

References

1. Marler, J. H., & Boudreau, J. W. (2017). *An evidence-based review of HR Analytics*. The International Journal of Human Resource Management, 28(1), 3–26.

2. Hom, P. W., Lee, T. W., Shaw, J. D., & Hausknecht, J. P. (2017). *One hundred years of employee turnover theory and research*. Journal of Applied Psychology, 102(3), 530–545.

3. U.S. Office of Personnel Management (OPM). *Federal Workforce Data*. https://data.opm.gov/